

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – X
PAPER –2
TEST DATE: 21-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

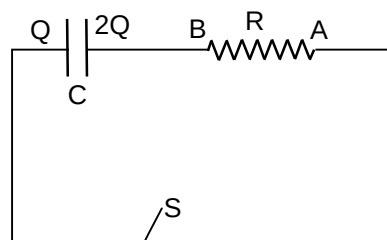
Physics

PART – I

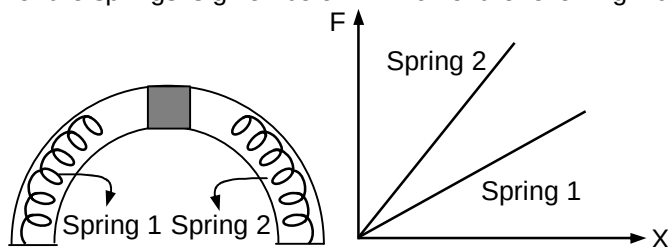
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Select correct alternatives if both plates of parallel plate capacitor are given positive charges Q and $2Q$ as shown in the figure and at $t = 0$, switch S is closed :

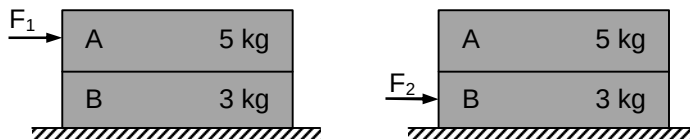


- (A) current in resistance 'R' will flow from A to B
- (B) electric field intensity between the plates of capacitor will decrease continuously
- (C) current in the circuit at time t is $\frac{Q}{2RC} e^{-t/RC}$
- (D) total heat dissipated in the circuit after having switch S closed is $\frac{Q^2}{4C}$
2. In a setup of displacement method experiment, the distance between the screen and a light source is 120 cm and the lens used has a small aperture. By moving the lens between the source and the screen, sharp images are obtained on the screen for two different positions of the lens. The ratio of sizes of these two images is 9 : 1. Then choose the correct option(s).
- (A) The focal length of the lens is 22.5 cm
- (B) The focal length of the lens is 45 cm
- (C) The separation between the two different positions of the lens is 30 cm
- (D) The separation between the two different positions of the lens is 60 cm
3. In the figure, a block rests on the top of a smooth fixed hemispherical tube of radius R in which it can just fit. Two springs are connected to the base as shown. The block is given a small jerk so that it can slide on the hemisphere. The F - X (F is magnitude of force and X is compression) graph for the springs is given below. Which of the following may be possible?



- (A) The block will compress both springs by same amount.
- (B) The block will compress the springs during its to and fro motion about its original position by different amounts.
- (C) The block will perform to and fro motion along the hemispherical surface about the original position.
- (D) The block can never come to the original position.

4. An ideal gas can be expanded from an initial state to a certain volume through two different processes (i) $PV^2 = \text{constant}$ and (ii) $P = KV^2$ where K is a positive constant. Then
 (A) final temperature in (i) will be greater than that in (ii)
 (B) final temperature in (ii) will be greater than that in (i)
 (C) total heat given to the gas in case (i) is greater than that in (ii)
 (D) total heat given to the gas in case (ii) is greater than that in (i)
5. A particle of $m = 1 \text{ kg}$ is moved in a circle of radius $R = 1 \text{ m}$ on a rough horizontal surface with coefficient of friction $\mu = 0.1$, with a constant speed of 1 m/s by an external force of constant magnitude. The magnitude of the external force is (Take $g = 10 \text{ m/s}^2$)
 (A) $\sqrt{2} \text{ N}$
 (B) $2\sqrt{2} \text{ N}$
 (C) $\sqrt{3} \text{ N}$
 (D) $2\sqrt{3} \text{ N}$
6. A block A of mass 5 kg rests over another block B of mass 3 kg which is placed on a smooth horizontal surface. There is friction between block A and block B. A horizontal force F_1 gradually increasing from zero to a maximum value is applied to A so that the blocks move together without relative motion. Instead of this, another horizontal force F_2 , gradually increasing from zero to a maximum value is applied to B so that the blocks move together without relative motion. Then
 (A) $F_1 (\text{max}) = F_2 (\text{max})$
 (B) $F_1 (\text{max}) > F_2 (\text{max})$
 (C) $F_1 (\text{max}) < F_2 (\text{max})$
 (D) $F_1 (\text{max}) : F_2 (\text{max}) = 5 : 3$



Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

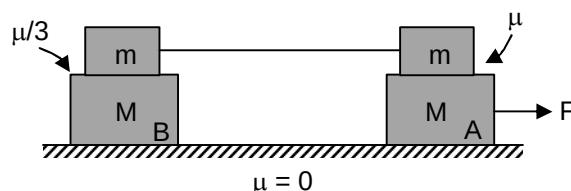
Paragraph for Question Nos. 07 and 08

A source emits sound waves of frequency 1000 Hz . The source moves to right with a speed of 32 m/s relative to ground. On the right there is a reflecting surface which moves to left with a speed of 64 m/s relative to ground. The speed of sound in air is 332 m/s .

7. The wavelength of sound in air emitted by the source is
 (A) 0.4 m
 (B) 0.22 m
 (C) 0.6 m
 (D) 0.33 m
8. The number of waves which meet the reflecting surface per second is
 (A) 1440
 (B) 1280
 (C) 1320
 (D) 1600

Paragraph for Question Nos. 09 and 10

In the figure shown, when a horizontal force F is applied to the larger block A, all the four blocks move together. There is no slipping between any two blocks. (Take $m = 1$ kg, $M = 2$ kg, $\mu = 0.3$ and $g = 10$ m/s²)

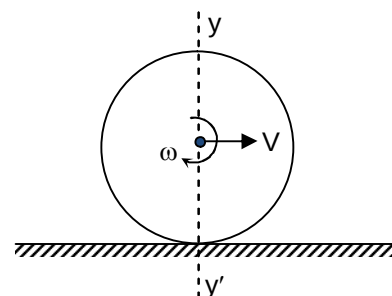
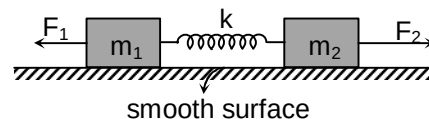


9. The maximum value of the applied horizontal force F is
 (A) 1 N
 (B) 2 N
 (C) 3 N
 (D) 4 N
10. In the above case, the friction force between the front blocks is
 (A) 1 N
 (B) 2 N
 (C) 3 N
 (D) 4 N

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A solid cylinder of mass M radius R is resting on a horizontal platform (which is parallel to x - y plane) with its axis along the y -axis. The platform is given a motion in X -direction given by $X = A \cos \omega t$. There is sufficient friction present in the surface of contact that can prevent the slipping between the cylinder and the platform. The maximum torque (in N-m) acting on the cylinder during its motion is (Take $M = 6$ kg, $R = 1$ m, $A = 2$ m, $\omega = 1$ rad/s)
12. Two blocks of masses $m_1 = 2$ kg and $m_2 = 4$ kg are connected by an ideal non-deformed spring of stiffness $k = 150$ N/m and placed on a smooth horizontal surface. Two horizontal forces $F_1 = 9$ N and $F_2 = 27$ N are applied on the blocks in the opposite directions as shown in the figure. Find the maximum elongation (in cm) produced in the spring.
13. A solid sphere of mass m and radius r is in pure rolling motion on a rough horizontal surface with linear speed v . The force exerted by the right half of section yy' on the left half of section yy' is $\frac{Kmv^2}{16r}$ newton. Find K .



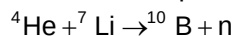
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

When an α -particle with kinetic energy $K = 11$ MeV collides with a stationary lithium nucleus, the following nuclear reaction takes place



The Q-value of this reaction is $E = -2.86$ MeV. If the kinetic energy of the neutron outgoing at right angle to the incoming direction of α -particle is X MeV and the kinetic energy of boron atom is Y MeV.

14. The value of X is.....

15. The value of Y is.....

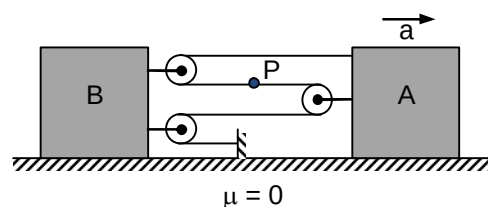
Question Stem for Question Nos. 16 and 17

Question Stem

In figure shown block A starts moving with acceleration a .

Magnitude of acceleration of block B is $\frac{Ma}{4}$ and

magnitude of acceleration of point P of string is $\frac{Ka}{2}$.



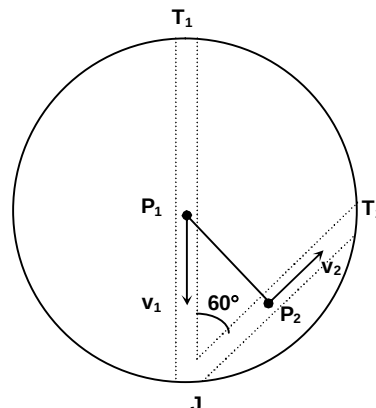
16. The value of M is.....

17. The value of K is.....

Question Stem for Question Nos. 18 and 19

Question Stem

Two tunnels are dug across the earth as shown in the figure. Two particles P_1 and P_2 are oscillating from one end to the other end of the tunnels T_1 and T_2 respectively. At some instant, particles are getting at the mid points of the tunnels as shown in the figure. Phase difference between the particles P_1 and P_2 is X° and the ratio of maximum velocities of particles P_1 and P_2 is Y .



18. The value of X is.....

19. The value of Y is.....

Chemistry

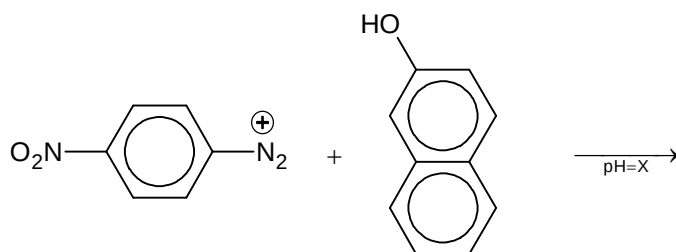
PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

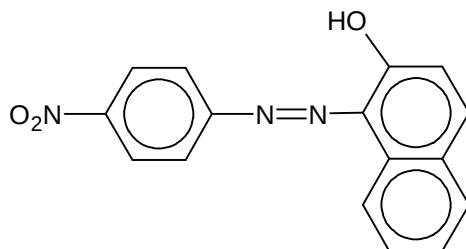
20. A non-reacting dry gas, (say N_2) is bubbled through some liquids separately and samples of emergent gas are collected at the same temperature and pressure. If definite volume of the emergent gas weighs more than the same volume of dry gas under identical conditions of temperature and pressure, the liquid(s) may be
- Heavy water (D_2O)
 - Methanol
 - Acetone
 - 1 : 1 molar mixture of H_2O and D_2O

21.



The incorrect statement (s) is/are

- 'X' should be greater than 7 so that phenolic group ($-OH$) convert into $\text{---}\overset{\ominus}{O}$ and it activate the aromatic ring towards electrophilic aromatic substitution
- 'X' should be less than 7 so that more positive charge present on N_2 and electrophilic power of electrophile increases in the reaction
- The above reaction is an example of electrophilic aromatic substitution
- If medium is basic in above reaction then product is



22. Select the correct statement(s) of the following:

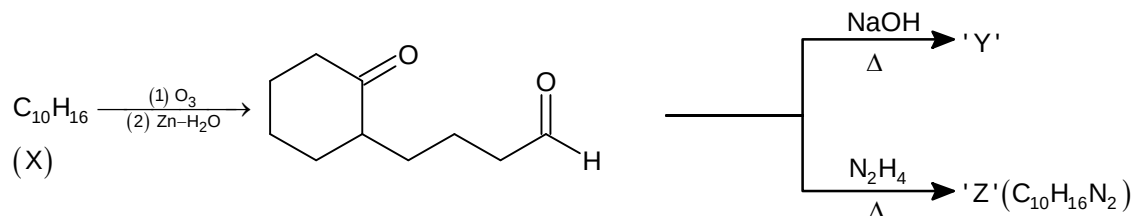
- If the critical micelle concentration of a soap is 10^{-3} M, then osmotic pressure of 1×10^{-4} M solution will be almost equal to that of 1×10^{-4} M NaCl solution
- When some oil soluble dye is added to water-in-oil emulsion, the background appears coloured
- In cataphoresis of a sol formed by peptization of freshly obtained SnO_2 by somewhat excess of NaOH, the sol particles move towards anode
- Coagulating power of the following electrolytes in the sol obtained in option (C) will be in the order $Na_3PO_4 > MgSO_4 > AlCl_3$

23. Which of the following statements is/are incorrect?
 (A) Entropy of a system is a measure of available energy to do the useful work.
 (B) In an irreversible process the entropy change between the two states is path dependent.
 (C) Complete conversion of heat into work is possible in a reversible isothermal expansion of ideal gas.
 (D) Decrease in free energy of a system represents the unavailable energy for doing useful work.
24. Which of the following statement is/are correct?
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2 - \text{O} - \text{CH}_3$ (Boiling point) .
 (B) p-nitrophenol is stronger acid than o-nitrophenol.
 (C) Phenol reacts with concentrated sulphuric acid and at low temperature major product is ortho sulfonated product.
 (D) Boiling point of ortho-nitrophenol is less than para nitrophenol.
25. Which of the following is/are alloy(s) containing copper?
 (A) Brass
 (B) Bellmetal
 (C) Nickeloy
 (D) Magnalium

Section – A (Maximum Marks: 12)

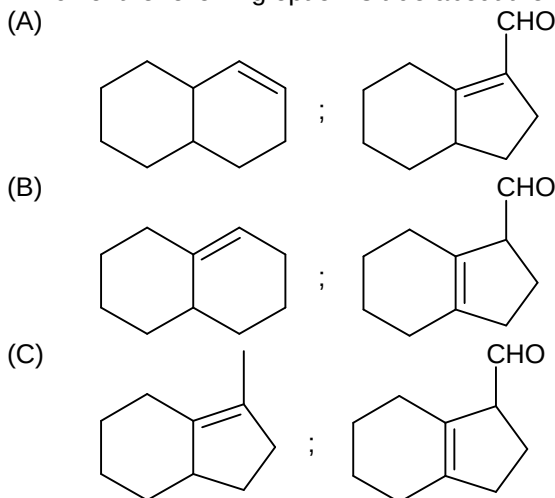
This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

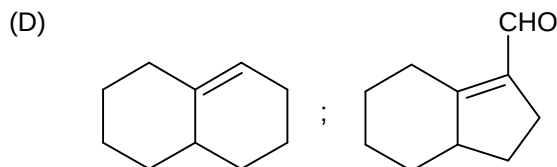
Paragraph for Question Nos. 26 and 27



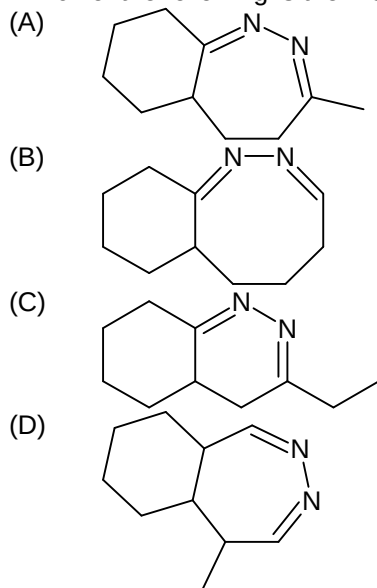
Considering the above sequence of reactions

26. Which of the following option is true about the structures of X and Y respectively?

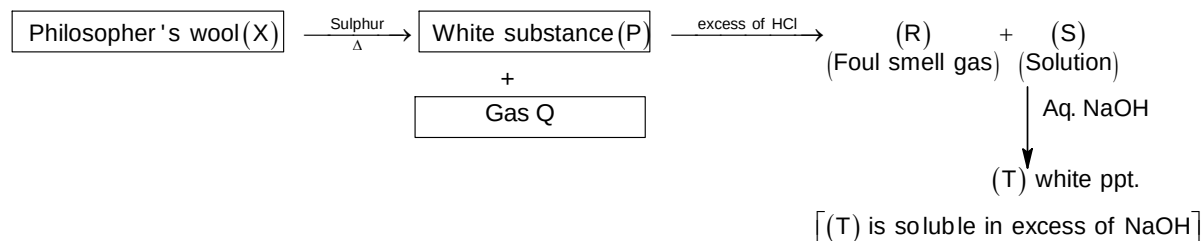




27. Which of the following is the most likely structure of 'Z'?



Paragraph for Question Nos. 28 and 29



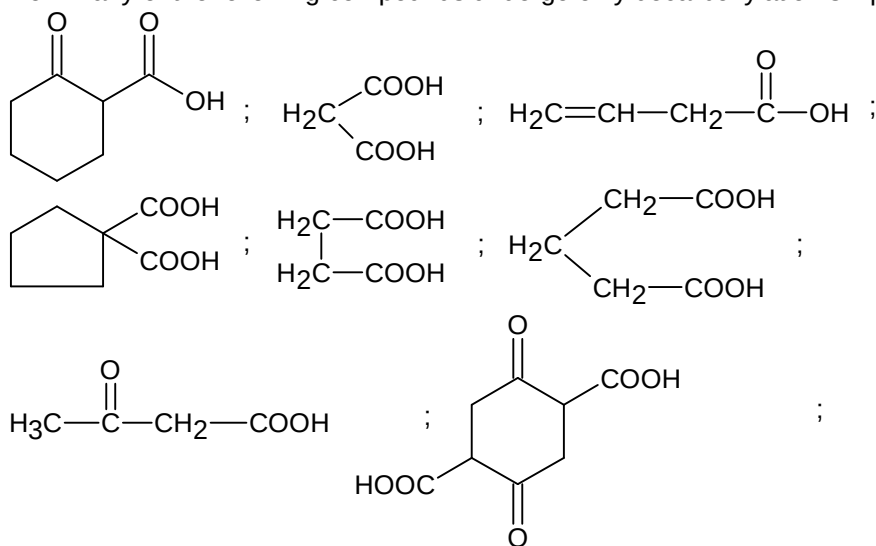
Considering the above sequence of reactions, answer the following.

28. Select the incorrect statement:
- Metal present in white substance (P) produces amphoteric oxide.
 - 'X' is yellow powder which on heating becomes white and again on cooling becomes yellow.
 - Mixture of (P) + BaSO_4 is called lithopone
 - 'X' on heating with cobalt nitrate gives Rinmann's Green.
29. Which of the following is not true characteristics of gas (R)
- It produces black precipitate, when passed through AgNO_3 solution.
 - It decolourises acidic solution of KMnO_4 .
 - It produces black precipitate, when passed through $(\text{CH}_3\text{COO})_2\text{Pb}$ solution.
 - It produces white precipitate, when passed through $(\text{CH}_3\text{COO})_2\text{Cd}$ solution.

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

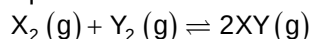
30. If the total number of compounds among the following, in which the difference in oxidation state of two atoms of same element (other than oxygen and hydrogen) is greater than 2 is 'P' and If the total number of compounds among following, in which oxidation state of two atoms of same element is same (other than oxygen and hydrogen) is 'Q'. Then what is the value of 'P' × 'Q'.
 $\text{Cr}_2\text{O}_7^{2-}$; NH_4NO_2 ; NH_4NO_3 ; Hydrazine; Calcium cyanamide; Urea; $\text{S}_2\text{O}_3^{2-}$; $\text{S}_4\text{O}_6^{2-}$; $\text{H}_2\text{S}_2\text{O}_5$
31. 3 mole of a mixture, having equal moles each of AgNO_3 ; NaNO_3 ; NH_4NO_3 ; LiNO_3 ; $\text{Ca}(\text{NO}_3)_2$; KNO_3 are heated strongly. What is total weight in gram of NO_2 (g) evolved?
32. How many of the following compounds undergo only decarboxylation simply on heating?

**Section – C (Maximum Marks: 12)**

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34**Question Stem**

A 2 litre flask containing a gas X_2 at a pressure of 1 atm is connected through a very narrow tube with a stop cock to another 3 litre flask containing a gas Y_2 at a pressure of 2 atm, the two flasks being at the same temperature. On opening the stopcock the two gases mix and react according to the following equation



At equilibrium the gaseous product XY has a partial pressure of 0.6 atm. (Assume negligible volume of the thin tube and no change in temperature of the flasks). Calculate the

33. Partial pressure of X_2 (in atm) at equilibrium.....

34. Equilibrium constant K_C of the reaction.....

Question Stem for Question Nos. 35 and 36

Question Stem

2.0 litre of 0.05 M KCN and 3.0 L of 0.05 M $M(CN)_2$ are mixed together at certain temperature T Kelvin, M being a strongly electropositive element. If ionization constant of CN^- is 2.0×10^{-4} and $[H^+]$ in pure water is 2.0×10^{-7} M at T Kelvin; If $X \times 10^{-4}$ is hydrolysis constant of CN^- ion and "Y" is pH of the resulting solution. Then,

35. What is the value of X?

36. What is the value of Y?

Question Stem for Question Nos. 37 and 38

Question Stem

When NO and NO_2 are mixed together, following equilibria were established in a closed vessel.



In an experiment, when NO and NO_2 are mixed in molar ratio of 1 : 2, final pressure was found to be 5.05 atm. When the partial pressure of N_2O_4 (at equilibrium) was 1.7 atm.

37. Calculate the equilibrium pressure of NO (in atm)?

38. What is the value of K_{P_2} (X) ?

Mathematics**PART – III****Section – A (Maximum Marks: 24)**

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let $A = \{1, 2, 3, \dots, 36\}$. Let B, C, D be set of all 1×1 , 2×2 , 3×3 matrices that can be formed using elements of A. If $|a|$ represents determinant of matrix 'a', then
- (A) $\sum_{a \in B} |a| = 666$
- (B) $\sum_{a \in C} |a| = 666^2$
- (C) $\sum_{a \in D} |a| = 666^3$
- (D) $\sum_{a \in C} |a| = \sum_{a \in D} |a|$
40. Let $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let C be a circle concentric with E and inscribed in E and S be locus of foot of perpendicular from foci of E to any tangent to E. If S is also locus of a point P such that chord of contact of tangents drawn from P to C subtends right angle at origin, then which of the following are true?
- (A) Eccentricity of E is less than $\frac{1}{2}$
- (B) Maximum distance between C and S is less than $\sqrt{2}$ times the radius of C
- (C) Maximum distance between C and E is greater than $\frac{2}{7}$ times the radius of C
- (D) S doesn't exist
41. Let $S(k) = 2022 - k$ and $P = \sum_{r=0}^{2022} (-1)^r {}^{2022}C_r [S(k)]^{2022}$, then
- (A) if $k = 0$, P is divisible by 43^{48}
- (B) if $k = 0$, P is divisible by 47^{43}
- (C) if $k = r$, P is divisible by 43^{48}
- (D) if $k = r$, P is divisible by 47^{43}
42. Let there be a column of square cross-section having side length 2 units and of infinite length centred on z-axis. Let P be a random point lying in the section cut off by the plane $x + 2y + 2z = 30$ in the column. Q is reflection of P in the plane $z = 15$ and R is locus of point Q, then
- (A) area of R is 6 square units
- (B) area of R is $\frac{12}{\sqrt{5}}$ square units
- (C) R lies in $x + 2y - 2z + 30 = 0$
- (D) R lies in $x - 2y - 2z + 30 = 0$

43. Let solutions of the equation $z^8 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ be represented by $\vec{a}_i : i = 1, 2, \dots, 8$ on Argand plane such that $\text{Arg}(a_i) < \text{Arg}(a_j) \forall i, j$ ($\text{Arg}(a) \in [0, 2\pi)$) (where $\text{Arg}(a)$ represents argument of $\vec{a}$), then
- (A) $\sum_{i=1}^8 \sum_{j=1}^8 \vec{a}_i \cdot \vec{a}_j = 0$
- (B) $\sum_{1 \leq i < j \leq 8} \vec{a}_i \cdot \vec{a}_j = -(4 + 4\sqrt{2})$
- (C) $\sum_{i=1}^8 \sum_{j=1}^8 (\vec{a}_i \times \vec{a}_j) = 0$
- (D) $\sum_{1 \leq i < j \leq 8} (\vec{a}_i \times \vec{a}_j) = 4 + 4\sqrt{2}$
44. A person walks at a uniform speed of 1 unit/min for a distance of 1 unit. He then turns right and covers $\frac{3}{4}$ of the distance he walked before turning and continues doing this. Let $P(t)$ represent the distance between the persons' current position and starting point $\forall t \geq 0$
- (A) $P(t)$ is a continuous function $\forall t$
- (B) $P(t)$ is a differentiable function $\forall t$
- (C) $\lim_{t \rightarrow \infty} P(t) = \frac{4}{5}$
- (D) $\lim_{t \rightarrow \infty} P(t) = \frac{16}{25}$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

Let p, q be selected from a set of first 20 natural numbers. Let $A = \left\{ \frac{r}{s} : r, s \in \{1, 2, \dots, 20\} \right\}$

45. If probability of $\frac{p}{q}$ being a natural number greater than 1 is k , then $10000k$ equals
- (A) 67
- (B) 66
- (C) 48
- (D) 46
46. If a random element of set A is selected, then probability that it is an integer is equal to $\frac{u}{v}$ (where u, v are co-prime integers), then $u + v$ is equals
- (A) 55
- (B) 275
- (C) 301
- (D) 321

Paragraph for Question Nos. 47 and 48

Let there be two particles of same mass inside an elliptical container placed along $\frac{x^2}{2} + y^2 = 1$. Initially, both particles are at (1, 0). At $t = 0$, first particles (P_1) starts moving towards negative x-axis and second particle (P_2) starts moving along a line at an angle of $\frac{\pi}{3}$ from positive x-axis, both having speed of $\sqrt{2}$ units/s. The particles can collide with container and with each other, all collisions being perfectly elastic. Let $f(t)$ be a function of distance between the two particles at any time t . Let $g(t)$ and $h(t)$ be function of distance of P_1 and P_2 respectively from (1, 0) at any time t

47. Let a, b, c be number of solutions of the equation $f(t) = 0, g(t) = 0$ and $h(t) = 0$ respectively $\forall t \in (0, 100]$, then
 (A) $a + b = 3c$
 (B) $a + c = 4b$
 (C) $b + c = 3a$
 (D) $a + b = 2c$
48. Which of the following are periodic functions?
 (A) $f(t)$
 (B) $g(t)$
 (C) $h(t)$
 (D) none of these

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. If in scalene $\triangle ABC$, $2c \sin(A - B) \cos(A + B) = (b \sin B - a \sin A)$ (where all symbols have their usual meanings), then $[5 \sin C]$ equals (where $[.]$ represents greatest integer function)
50. Number of distinct quadratic equations with real roots such that the equation remains unchanged if their roots are cubed equals
51. If the minimum value of the expression $\frac{x^2y^2 - 2xy^2 + 2y^2 + 4xy - 4y + 4}{xy^2 + 2y}$ is ' k ' $\forall x, y, \in \mathbb{R}^+$, then $[10k]$ equals (where $[.]$ represents greatest integer function)

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $I_1 = \int_0^\infty \frac{x-1}{\sqrt{e^x-1} \ln(e^x-1)} dx$ and $I_2 = \int_0^4 \frac{\ln x dx}{\sqrt{4x-x^2}}$

52. The value of I_1 equals
53. The value of I_2 equals

Question Stem for Question Nos. 54 and 55

Question Stem

Let z_1, z_2, z_3 be three complex numbers such that $z_1 + z_2 + z_3 = i$, $z_1z_2 + z_2z_3 + z_1z_3 = -1$, $z_1z_2z_3 = -i$
 Let maximum value of $|z_1 - kz_2 + (k - 1)z_3| \forall k \in [0, 1]$ equals 'a' and minimum value of $|z - z_1|^2 + |z - z_2|^2 + |z - z_3|^2 \forall z$ is 'b' (where $i = \sqrt{-1}$)

54. The value of 'a' is

55. The value of 'b' is

Question Stem for Question Nos. 56 and 57

Question Stem

Let L_1, L_2, L_3 be three straight lines such that no two of them are parallel. Let C_1, C_2, C_3, C_4 be circles touching all the three lines such that radius of C_1 is smallest. Let P be point of concurrence of the altitudes of the triangle formed by centres of C_2, C_3 and C_4 . If L_1, L_2, L_3 intersect each other at points A, B and C such that $AB = 3$, $BC = 4$ and $\angle ABC = \frac{\pi}{2}$. Let $m = \min\{PA, PB, PC\}$; $n = \max\{PA, PB, PC\}$

56. The value of m equals

57. The value of n equals